ANTIMICROBIAL MANAGEMENT OF CYSTIC FIBROSIS EXACERBATIONS

I. OVERVIEW: Cystic fibrosis (CF) is a genetic condition that affects multiple organ systems, most notably resulting in chronic airway infection and progressive lung disease marked by intermittent exacerbations of respiratory symptoms. As a result, people with CF (pwCF) require periodic admission to the hospital for management, including antimicrobial therapy. Over time, bacteria infecting the airways become resistant to antibiotics due to cumulative antibiotic exposure. This complicates the antibiotic management of pulmonary exacerbations.

II. PURPOSE: The purpose of this document is to provide guidance to inpatient medical providers who are caring for pwCF admitted for management of a pulmonary exacerbation requiring the use of antimicrobials (antibiotics or antifungals). Current practices are variable, and while some variability is necessary due to the complexities of each unique patient, this guideline will provide recommendations to standardize the care of these patients whenever possible, as well as streamline the process for approval of restricted antimicrobials.

III. SCOPE: This guideline includes recommendations for the antimicrobial management of non-neonatal pwCF (90 days and older) who are admitted to a pediatric inpatient general care (non-ICU) service for treatment of an acute pulmonary exacerbation. This guideline is not intended for use in patients who are post-lung transplant or who are critically ill; while aspects of this guideline could be applied to those situations, the care of these patient populations requires additional consideration beyond the scope of this guideline.

This guideline includes recommendations for antimicrobial therapy based on patient factors and available culture data. These guidelines do not address long-term antimicrobial treatment or treatment for CF exacerbations that are managed exclusively in the outpatient setting. These guidelines also do not address the other essential aspects of treatment for a pulmonary exacerbation, such as airway clearance. Pediatric Infectious Diseases (ID) or other relevant consultation may be of benefit for patients who fall outside of the scope of these guidelines.

IV. DEFINITIONS:

- Pulmonary exacerbation: There is a lack of consensus on the definition of a CF pulmonary exacerbation, although these events are generally characterized by an intermittent worsening of respiratory symptoms (such as increased cough, sputum production, shortness of breath), often associated with a decrease in pulmonary function, weight loss, and other systemic symptoms.
- Restricted Antimicrobials
    i. The use of some antimicrobials is restricted in all hospitalized patients, based on criteria developed by the Antimicrobial Subcommittee and approved by the Pharmacy and Therapeutics Committee
    ii. **Tier 1:** Tier 1 antimicrobials require prior approval from the pediatric antimicrobial stewardship program (ASP) or Pediatric ID consult team.
        1. Tier 1 antimicrobials that are commonly used for the treatment of CF exacerbations include carbapenems (e.g., meropenem), new beta-lactam/beta-lactamase inhibitor combinations (e.g., ceftazidime/avibactam or meropenem/vaborbactam), linezolid, and voriconazole.
    iii. **Tier 2:** Tier 2 antimicrobials may be initiated without obtaining prior approval but are reviewed for appropriateness by the ASP within 24-72 hours of initiation and need to either meet certain pre- existing criteria or require approval from the pediatric antimicrobial stewardship program (ASP) or Pediatric ID consult team for continued use.
        1. Tier 2 antimicrobials that are commonly used for the treatment of CF exacerbations include cefepime, ceftazidime, piperacillin/tazobactam, aztreonam, fluoroquinolones (e.g., ciprofloxacin), and vancomycin.

V. GUIDELINE

Initial Antimicrobial Selection

- A critically ill pwCF (severe sepsis or severe respiratory failure) falls outside the scope of this guideline, but the antimicrobial regimen should, at a minimum, include coverage for all bacterial species grown from respiratory culture within the last year, as well as empiric coverage for *Staphylococcus aureus*, *Pseudomonas aeruginosa*, and *Burkholderia cepacia* complex.
    i. Pediatric Pulmonology and Pediatric Infectious Diseases should be consulted to further discuss the appropriate treatment for critically ill patients
- Otherwise, the choice of antimicrobials should be based on consideration of the following:
    i. Typical airway pathogens for age
        1. In young children with CF, the primary airway pathogens are oral commensal bacteria and *S. aureus*; most children will not require anti-pseudomonal antibiotics if they have not had a respiratory culture grow *P. aeruginosa* within the past 2 years.
    ii. Recent respiratory culture data (coverage for bacterial species grown from respiratory cultures within the past 1 year, or past 2 years for *P. aeruginosa*)
        1. An anti-pseudomonal antibiotic is not necessary for initial therapy unless patients have had:
            a. Respiratory culture positive for *P. aeruginosa* in the past 2 years
            b. Respiratory culture positive for non-aeruginosa *Pseudomonas* species (*Pseudomonas putida, Pseudomonas fluorescens*, etc.) in the past 1 year
        2. Culture results from a patient's sibling should not be used to guide routine treatment, but may be considered in the setting of critical illness
        3. See "Incorporating P. aeruginosa susceptibility data into treatment decisions" below for more details on use of susceptibility data for P. aeruginosa
    iii. Recent outpatient antimicrobial use
    iv. Previous antibiotic regimens that have resulted in clinical improvement
        1. If a patient has had clinical improvement with a non-preferred agent for P. aeruginosa (such as meropenem) based on lack of improvement with preferred agents in the past, a repeat trial of a preferred agent should be considered with subsequent exacerbations
        2. If a patient has had clinical improvement with agents directed at bacteria not found in their respiratory cultures (e.g., improvement with antipseudomonals in the absence of P. aeruginosa identification on culture), a repeat trial of culture-directed agents (e.g., with non-antipseudomonals) should be considered with subsequent exacerbations
    v. Patient allergies and adverse effects
        1. For patients with penicillin or cephalosporin allergy, refer to β-lactam Allergy Evaluation and Empiric Therapy Guidance for alternative β-lactam treatment options; consultation to the β-lactam Allergy Evaluation Service (BLAES) is recommended.
        2. If there is a history of other complications or adverse effects (e.g., hearing loss due to aminoglycoside use), alternatives to that antibiotic(s) should be considered
- Incorporating P. aeruginosa susceptibility data into treatment decisions:
    i. Treatment with antibiotics based on susceptibility testing results for P. aeruginosa does not consistently correlate with clinical outcomes in pwCF
        1. Changes to an antimicrobial regimen should not be made based on new susceptibility testing results for P. aeruginosa if the patient is doing well on the current regimen
        2. If there has been no improvement with the initially selected antimicrobial regimen, consider changes based on results of susceptibility testing for P. aeruginosa

Initial Antibiotic Recommendations by Organism

- Oral microbes only (including *Moraxella*, *H. influenzae*):
  - i. PREFERRED: **Ampicillin/sulbactam**
  - **ii. BETA-LACTAM ALLERGY:**
    1. *Consultation to the β-lactam Allergy Evaluation Service (BLAES) for allergy assessment is recommended if not previously done.*
    2. For Low/High-risk (See Figure 1**)** Penicillin Allergy: **Ceftriaxone**
    3. For High-risk (See Figure 1**)** Cephalosporin Allergy: **Levofloxacin** or **Doxycycline**
    4. For Severe Non-IgE mediated (See Figure 1) Allergy: **Levofloxacin** or **Doxycycline**
- *P. aeruginosa*
  - i. There are few data to support the practice of using two antibiotics with anti-pseudomonal activity; however, this is standard practice currently supported by the CF Foundation
  - ii. **PREFERRED: Cefepime** PLUS **IV tobramycin** or a PO fluoroquinolone (**ciprofloxacin** or **levofloxacin**)
    1. **Ceftazidime** or **piperacillin/tazobactam** are preferred alternatives to cefepime based on individual patient factors
    2. Broader-spectrum agents such as carbapenems should be considered only if the above agents have been ineffective previously, or cannot be used due to allergy or adverse effects
  - iii. **FOR BETA-LACTAM ALLERGY:**
    1. *Consultation to the β-lactam Allergy Evaluation Service (BLAES) for allergy assessment is recommended if not previously done.*
    2. For Low/High-risk (See Figure 1) Penicillin Allergy: **Cefepime**
    3. For Low-risk (See Figure 1) Cephalosporin Allergy: **Cefepime**
       - *a. unless the reaction was w/ Ceftriaxone, Cefotaxime or Cefpodoxime then: Ceftazidime*
    4. For High-risk (See Figure 1) Cephalosporin allergy: **Meropenem**
    5. For Severe Non-IgE mediated (See Figure 1) Allergy: **Aztreonam**
- Non-aeruginosa Pseudomonas species (e.g., *P. fluorescens*, others)
  - i. Double coverage of Pseudomonas species other than P. aeruginosa is not indicated
  - ii. **PREFERRED**: **Cefepime**
    1. **Ceftazidime** or **piperacillin/tazobactam** are preferred alternatives to cefepime based on individual patient factors
    2. Broader-spectrum agents such as carbapenems should be considered only if the above agents have been ineffective previously, or cannot be used due to allergy or adverse effects
  - iii. **FOR BETA-LACTAM ALLERGY**
    1. *Consultation to the β-lactam Allergy Evaluation Service (BLAES) for allergy assessment is recommended if not previously done.*
    2. For Low/High-risk (See Figure 1) Penicillin Allergy: **Cefepime**
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       - *a. unless the reaction was w/ Ceftriaxone, Cefotaxime or Cefpodoxime then: Ceftazidime*
    4. For High-risk (See Figure 1) Cephalosporin Allergy: **Meropenem**
    5. For Severe Non-IgE mediated (See Figure 1) Allergy: **Aztreonam**
- *S. aureus:*
  - i. Double coverage of S. aureus is not indicated
  - ii. Methicillin susceptible (MSSA)
    1. **PREFERRED: Cefazolin**
    2. If other agents with activity against MSSA (such as **cefuroxime, cefotaxime, cefepime, ampicillin-sulbactam, piperacillin/tazobactam**) are being used for treatment of other organisms, additional antibiotics are not needed for MSSA

3. Other options include oral, **trimethoprim/sulfamethoxazole (TMP-SMX), doxycycline, linezolid, clindamycin (if sensitive)**
    a. The above antibiotics are highly bioavailable and there is no benefit when given IV, if a patient can tolerate the oral form
4. Avoid vancomycin whenever possible

iii. Methicillin resistant (MRSA)
1. **PREFERRED:** Oral **trimethoprim/sulfamethoxazole (TMP-SMX), doxycycline, linezolid, clindamycin (if sensitive)**
2. For hospitalized patients, it is reasonable to first try one of the above highly bioavailable oral options, but if a patient has failed these options, **vancomycin** is the preferred IV agent

- Other Gram-negative rods (GNRs):
    i. Susceptibility results may be considered when selecting antibiotics for GNR other than *P. aeruginosa*
    ii. Use of two agents (double coverage) for GNR other than *P. aeruginosa or Burkholderia* species is not indicated
    iii. ***Achromobacter species***
        1. PREFERRED: **Piperacillin/tazobactam or ceftazidime;**
        2. ALTERNATIVES: **Oral TMP-SMX or minocycline**
    iv. ***Acinetobacter* species (including *A. baumanni* complex)**
        1. PREFERRED: **Cefepime**
        2. ALTERNATIVES: **Ampicillin/sulbactam, oral TMP-SMX (if susceptible) Levofloxacin** or **minocycline**
    v. ***Burkholderia* species** *(at least two active agents are preferred)*
        1. PREFERRED: **TMP-SMX**
        2. ALTERNATIVES: **Oral Minocycline, Meropenem, Ceftazidime** or **levofloxacin**
    vi. ***Stenotrophomonas maltophilia* complex**
        1. PREFERRED: TMP-SMX
        2. ALTERNATIVES: Oral levofloxacin, or minocycline
- Non-tuberculous mycobacteria (NTM):
    i. If there is concern that NTM is contributing to current symptoms and/or lung function decline, consult Pediatric ID or refer to Pulmonary-ID combined clinic
        1. In conjunction with Pediatric ID consult team, consider treatment for NTM pulmonary disease if there is recurrent growth of the same NTM species from respiratory culture, lung imaging consistent with NTM disease, and no clinical improvement despite treatment with antibiotics directed at other bacteria from airway cultures, and optimization of all other therapies and comorbidities (airway clearance, nutrition, etc.)
    ii. Many drug-drug interactions exist with antimicrobials used for NTM in pwCF (e.g., rifampin, macrolides, fluoroquinolones, linezolid, SSRIs, CFTR modulators). Careful evaluation of drug interactions must occur when making treatment plans for NTM. Consult with CF PharmD for evaluation for potential dose reduction of CFTR modulators when treating NTM pulmonary disease.
- Aspergillus species
    i. Aspergillus may contribute to pulmonary symptoms in multiple ways.
        1. If concern for Allergic Bronchopulmonary Aspergillosis (ABPA), follow existing CF Foundation treatment guidelines.
        2. If invasive Aspergillus is suspected, in conjunction with Pediatric ID consult team, consider treatment if there is recurrent growth of Aspergillus from respiratory culture, lung imaging consistent with Aspergillus disease, and no clinical improvement despite treatment with antibiotics and optimization of all other therapies and comorbidities (airway clearance, nutrition, etc.). Refer to **Guidelines for Treatment of Invasive Aspergillus**.

ii. Consult with CF PharmD for evaluation for potential dose reduction of CFTR modulators when treating Aspergillus (**See Table 2**).
iii. Other molds (e.g., *Exophiala, Scedosporium/Lomentospora, Penicillium*, others) rarely require treatment. Consult Pediatric ID to discuss potential treatment if indicated

Approval for restricted antimicrobials:

- If a tier 1 restricted antimicrobial is indicated based on the above guideline or other patient factors, the Pediatric Antimicrobial Stewardship team (pager 36149) should be contacted (prior to 5pm on the day of planned admissions) and approval noted in admission documentation, in the following manner:
    i. Direct admit from clinic or home: pediatric CF clinical pharmacy specialist will call Pediatric Antimicrobial Stewardship team
        1. When pediatric CF clinical pharmacy specialist is out, admitting provider will call Pediatric Antimicrobial Stewardship team
    ii. Admit from ED: Admitting pulmonology fellow or attending will call Pediatric ASP team
    iii. Notation in admission documentation should state information such as, "Tier 1 restricted antimicrobial reviewed and approved by [name of Pediatric Antimicrobial Stewardship team member] on [date/time]."
- If a tier 2 restricted antimicrobial is indicated based on the above guideline or other patient factors, the primary team does NOT need prior approval, but should ensure they meet criteria for continued use or have another justification for its use.

Oral Antibiotics:

- For highly bioavailable antibiotics (**See Table 1**), the oral route should be utilized if the patient is able to tolerate enteral medications

Home antibiotics (including inhaled antibiotics):

- Home azithromycin should be continued unless there are concerns about side effects such as QT prolongation with other medications being used (such as fluoroquinolones)
- Home inhaled antibiotics are generally continued
- Inhaled tobramycin and colistimethate are on hospital formulary. Inhaled aztreonam is not formulary.

Re-evaluation of antimicrobials based on new culture results

- If a new bacterial species grows in culture after admission, the initially selected antibiotic regimen may be continued without changes if the patient is improving (with the exception of *P. aeruginosa* and *Burkholderia* species, which should always be treated if isolated from culture)
    i. Refer to Eradication of Initial *P. aeruginosa* Clinical Care Guidelines
- If the patient is not improving on the initial antibiotic regimen, it is reasonable to change antibiotics to cover the new bacterial species
- Narrowing antibiotics based on lack of growth may be appropriate in some circumstances and should be considered on a case-by-case basis
- Antibiotics should not be adjusted based solely on new susceptibility results for P. aeruginosa (see **"Incorporating *P. aeruginosa* susceptibility data into treatment decisions"** above), unless the patient is not improving on the current antibiotic regimen

Re-evaluation of antimicrobials after 5-7 days of therapy

- If patients remain admitted, re-evaluate at approximately day 5-7 of antimicrobial therapy, which may include review of clinical status, respiratory support, and repeat PFTs
    i. If patient is not improving:
        1. Reassess other contributing factors (such as airway clearance, co-morbidities)

2. Consider further evaluation (such as bronchoscopy or chest CT), if appropriate
3. Consider Pediatric ID consult to discuss changes to antimicrobial regimen

Pediatric ID Consultation:

- Consider Pediatric ID consultation in the following scenarios:
  i. Difficulty choosing an empiric regimen due to multi-drug resistant organisms and/or multiple allergies/intolerances
  ii. Growth of an unfamiliar organism, or one of unclear significance including NTM, yeast (not Candida species), or molds
  iii. Discussion of new antibiotic regimen after failure to improve on the chosen empiric regimen after 5-7 days

Duration of antimicrobials

- Antimicrobials should be continued for a 10-14 day course in most cases (recent evidence from the STOP2 trial suggests that 21 days of therapy for exacerbation is not superior to 14 days). If all or almost all 14 days are completed in the hospital, additional antibiotics at home are not indicated

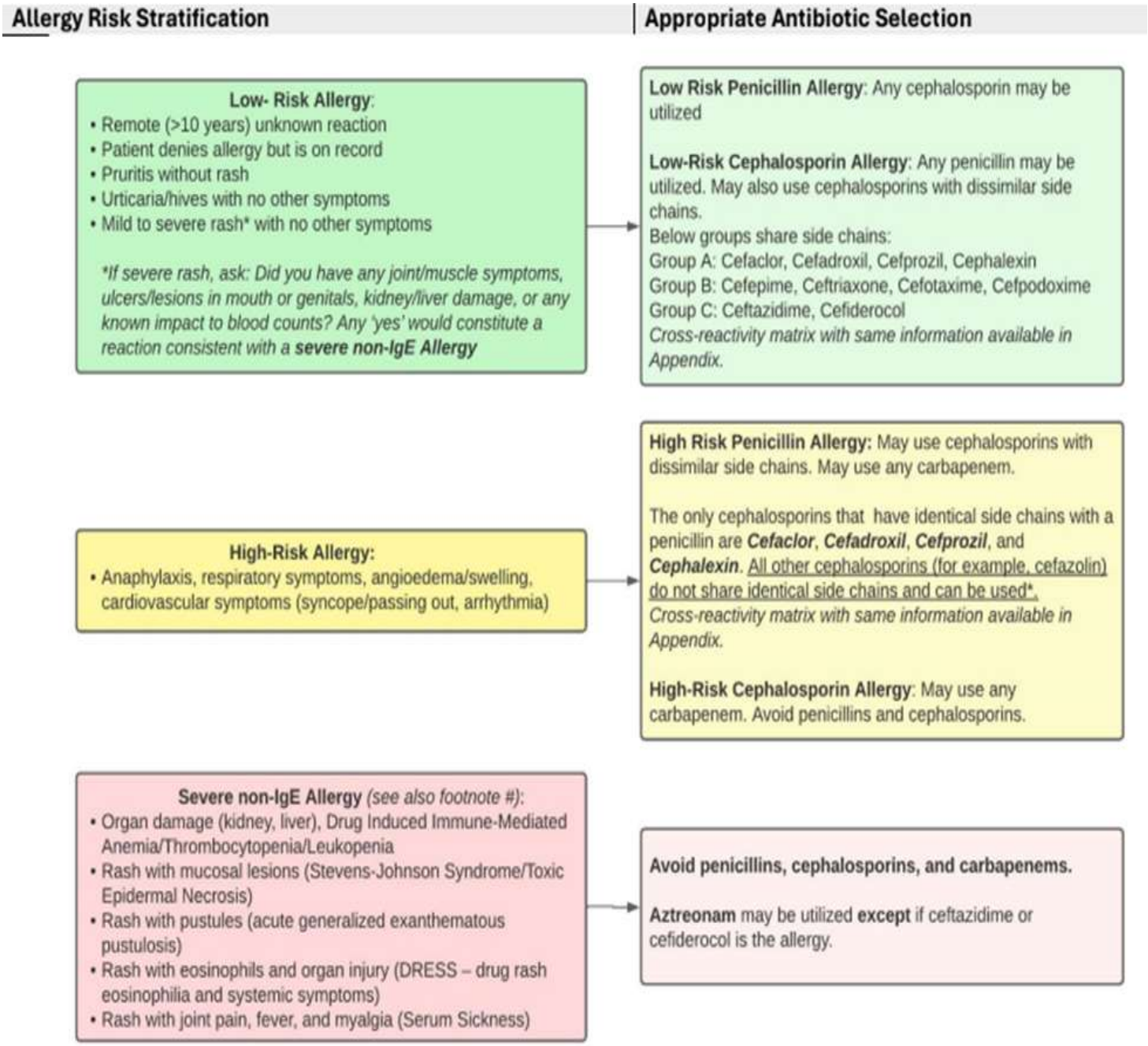
Antimicrobial Dosing and Monitoring

- Dosing for Antimicrobials in CF **(See "Antimicrobial Dosing Recommendations for Pediatric Patients")**
- Routine monitoring of antimicrobials in CF (**See Table 1**).

Discharge Planning

- **If patient is planned to go home with IV antibiotics to complete course,** inpatient pharmacist specialist to ask resident assistant to schedule patient follow up appointment with CF Pharmacist, by contacting pediatric pulmonary scheduling and requesting a CF pharmacist video visit for home IV antibiotic follow.
  i. This video visit should take place approximately 2-3 business days before PICC line is to be removed.
  ii. If visit cannot be scheduled due to lack of available appointments, resident assistant to message CF Pharmacist pool in Michart (MH PED PUL CF PHARM D) to inform of inability to schedule.
- **If patient received IV aminoglycoside (tobramycin or amikacin) as part of antibiotic course,** inpatient pharmacist specialist to ask resident assistant to schedule audiology screening appt with pediatric audiology (i.e., patient should have an appointment scheduled/set up before leaving), screening to take place within 3 months following discharge.

Figure 1. Beta-Lactam Allergy Assessment:



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Table 1. Antimicrobials for Pulmonary Exacerbations in Cystic Fibrosis Patients:

Antimicrobial Agent	Inpatient Monitoring Minimum Frequency (# times/week)				Antimicrobial Dosing	Enteral Preferred
	CBC-Diff	BMP	ALT, AST, ALK, Tbili	Audiology Screening		
Amikacin[1]	1 (IV only)	2 (IV only)	--	See Table 3	https://antimicrobialstewardship.med.umich.edu/guidelines/pediatric/antimicrobial-dosing-pediatrics	--
Ampicillin +/- sulbactam	1	1	+	--		--
Aztreonam (IV only)	1	1	+	--		--
Azithromycin (IV only)	1	--	1	--		PO
Cefazolin	1	1	--	--		--
Cefepime	1	1	--	--		--
Cefoxitin	1	1	+	--		--
Ceftazidime +/- avibactam	1	1	--	--		--
Ceftolozane-tazobactam	1	1	+	--		--
Ceftriaxone	1	+	+	--		--
Ceftaroline	1	1	--	--		--
Cefiderocol	1	1	--	--		--
Ciprofloxacin[3]	1	1	+	--		PO
Clindamycin	+	+	+	--		PO
Doxycycline	+	+	+	--		PO
Imipenem-cilstatin +/- relebactam	1	1	+	--		--
Levofloxacin[3]	1	1	+	--		PO
Linezolid[2]	1	--	1	--		PO
Minocycline	+	+	1	--		
Meropenem +/- vaborbactam	1	1	+	--		--
Omadacycline	+	+	+	--		PO
Piperacillin/tazobactam	1	1	+	--		--
Trimethoprim/sulfamethoxazole	1	1	1	--		PO
Tobramycin[1]	1 (IV only)	2 (IV only)	--	See Table 3		--
Vancomycin (IV only)[1]	1	2	--	--		--
Voriconazole[2,3]	1	1 (IV only)	1	--		PO
Posaconazole[2,3]	1	1 (IV only)	1	--		PO (DR tab, *NOT suspension*)
Itraconazole[2,3]	1	--	1	--		PO (Suspension, avoid cap)
Isavuconazole[2,3]	1	1 (IV only)	1	--		PO

Footnotes:

1. *Therapeutic Drug Monitoring may be indicated after pharmacist consultation*
2. *Therapeutic Drug Monitoring may be indicated after consultation with infectious disease pharmacist (pg: 36149)*
3. *Consider baseline EKG if receiving multiple medications known to prolong QT interval*

+. *At baseline and once again if therapy continues beyond 14 days*

Table 2. Azole Antifungal and CFTR Modulator Drug Interaction Management

CFTR Modulators	Drug Interaction w/ Azole Antifungals				Management During Azole Therapy
	Voriconazole	Posaconazole	Itraconazole	Isavuconazole	
Kalydeco© (Ivacaftor)	↑ Ivacaftor	↑ Ivacaftor	↑ Ivacaftor	↑ Ivacaftor	Decrease ivacaftor dose to 2x weekly, approximately 3 to 4 days apart
Orkambi© (Lumacaftor/Ivacaftor)	↓Voriconazole ↑ Ivacaftor ↑ Lumacaftor	↓Posaconazole ↑ Ivacaftor ↑ Lumacaftor	↓Itraconazole ↑ Ivacaftor ↑ Lumacaftor	↓Isavuconazole ↑ Ivacaftor ↑ Lumacaftor	Consider holding lumacaftor/ivacaftor during therapy. Azole is unlikely to achieve adequate concentrations.
Symdeko© (Tezacaftor/Ivacaftor)	↑ Tezacaftor ↑ Ivacaftor	↑ Tezacaftor ↑ Ivacaftor	↑ Tezacaftor ↑ Ivacaftor	↑ Tezacaftor ↑ Ivacaftor	Decrease dosing to: give/take morning tezacaftor/ivacaftor dose to 2x weekly, approximately 3 to 4 days apart. Omit evening ivacaftor dose.
Trikafta© (Elexacaftor/Tezacaftor/Ivacaftor)	↑ Elexacaftor ↑ Tezacaftor ↑ Ivacaftor	↑ Elexacaftor ↑ Tezacaftor ↑ Ivacaftor	↑ Elexacaftor ↑ Tezacaftor ↑ Ivacaftor	↑ Elexacaftor ↑ Tezacaftor ↑ Ivacaftor	Decrease dosing to: give/take morning elexacaftor/tezacaftor/ivacaftor dose to 2x weekly, approximately 3 to 4 days apart. Omit evening ivacaftor dose.

Table 3. UM Pediatric CF Audiology Screening Algorithm

Ototoxic Medication Regimen	Recommended Monitoring
Chronic* Inhaled Aminoglycoside for *Pseudomonas*	• Baseline screening with audio ototoxic monitoring (30-60 min test) • Annual screening – audio ototoxic monitoring (30-60 min test) via audiology paired/coordinated with a CF clinic visit
For EACH IV Aminoglycoside Course	• Screening with audio ototoxic monitoring (30-60 min test) within 3 months after completion of IV course. Scheduling for screening should be done prior to discharge. • Subsequent vestibular or audiology testing and OTO clinic visits dependent on findings, determined by Audiology.
Amikacin for NTM treatment • Per CFF/ESCF NTM guidelines: ○ IV amikacin up to 3 months duration ○ Inhaled amikacin for 12 months beyond culture conversion**	• Baseline screening with audio ototoxic monitoring (30-60 min test), before starting therapy • During IV amikacin – monthly screening, audio ototoxic monitoring (30 min test) • During Inhaled amikacin – every 2-4 months, audio ototoxic monitoring (30 min test)

**Defined as 2 or more consecutive cycled courses of inhaled aminoglycoside*

***Defined as three consecutive negative cultures, with the time of conversion being the date of the first of the three negative cultures, as long as no positive cultures are obtained during those 12 months*

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The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.

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